

Shrinking body size may not provide meaningful thermoregulatory benefits in a warmer world

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In a recent and impressive analysis of avian morphological data (covering >250,000 male birds from 105 species), Youngflesh et al.¹ report that birds breeding in North America have become significantly lighter over the past three decades, coincident with rising breeding season temperatures. Because these observations recapitulate predictions under Bergmann's rule (that is, that the body size of congeners and conspecifics is usually larger near the cooler poles than near the warmer Equator), the authors argue a thermoregulatory benefit to their occurrence under the assumption that smaller-bodied animals have lower cooling costs than larger-bodied animals in a warming world. We agree that warmer environments during reproduction may well explain avian body size declines. However, we question whether (1) changes in heat balance attributed to these declines are sufficiently large to explain observed size reductions, and (2) increased thermoregulatory efficiency during the short windows where reproduction occurs is always relevant, particularly when tenancy in breeding-ground temperatures is short (that is, among migrants). Generalizability of these results may be limited further by sex-specific enquiry, which overlooks the possibility of divergent selection on body size in males and females under climate warming. In this commentary, we expand on these concerns hoping to instigate discussion on knowledge gaps that need closing if we wish to better understand the proximate and ultimate drivers of shapeshifting animals in a changing world.

Data presented by Youngflesh et al.¹ corroborate several recent reports detailing how animals are changing body size and shape in parallel with warming temperatures^{2–4}. These collective observations are notable, and we share the view of Youngflesh et al.¹ that such temporal trends must be described and understood from both evolutionary and applied points of view^{5,6}. We also agree that smaller animals are theoretically better equipped to withstand higher ambient temperatures in thermal environments characterized by net heat loss (that is, where the animal remains warmer than its surroundings), since their higher surface area to volume ratios increase the proportion of integument usable for heat transfer. However, the reported mean body size shifts in this and other studies are small (here, -0.56% ; Fig. 1a) and so, according to currently available allometric relationships, yield

near-negligible effects on mean body surface area (-0.38%) and surface area to volume ratios ($+0.19\%$) across species (Fig. 1b). Such minor changes in morphology hold little thermoregulatory bearing, with conservative heat balance models (that is, assuming both complete and static plumage cover) suggesting a mean increase in heat flux (W cm^{-2}) of around 0.14% (Fig. 1b). The estimated effects on evaporative water loss ($+0.10\%$), total heat loss rate (that is, thermal conductance; $+0.27\%$), basal heat production (that is, basal metabolic rate; -0.39%), upper critical temperature (-0.05%) and the metabolic response to warmth (-0.48%) are similarly weak (Fig. 1b,c). Even for higher estimates of body mass change (for example, -2.6% on average in ref. 4), thermoregulatory benefits are still likely to be limited (for example, $+0.60\%$ in heat flux, $+0.42\%$ in evaporative water loss, $+1.10\%$ in thermal conductance and -1.56% in basal metabolic rate). By comparison, a typical physiological thermoregulatory response may lead to a 5- to 15-fold increase in evaporative water loss upon heat exposure⁷. In all, it seems doubtful that the observed shifts in body mass or size reported by Youngflesh et al.¹ and elsewhere will hold any significant bearing over the degree to which species can respond to thermoregulatory pressures caused by warming temperatures. In fact, available data suggest that, interspecifically, maximum tolerable air temperature in passerines has positive allometry⁷, and reducing plumage depth by as little as 1% would over-double the heat dissipation effect ascribed to body shrinkage in ref. 1 (that is, 0.39% reduction in heat flux relative to 0.14%). We therefore caution against such adaptationist conclusions in the absence of empirical evidence of thermoregulatory benefits.

Second, the authors base inference on data obtained during Northern Hemisphere springs and summers (1 May to 31 July). While this in itself is perfectly reasonable, the overwhelming majority of the species studied were migrants (66%, or 69 of 105 species) or partial migrants (22%, 23 of 105)⁸, suggesting short tenancy in at least some study regions, or even measurement during migratory stop-over. In these contexts, regressing morphology against local ambient temperature appears problematic. Specifically, conceptualization and hypothesis testing become difficult when the precise temperatures to which the birds are responding to morphologically are unknown,

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